

3. Simple harmonic motion (SHM) may be represented by:

$$a = -\omega^2 x$$

(a) State what quantities are represented by: [1]

a

ω

x

(b) If the amplitude of the oscillation of an object moving with SHM is 0.012m and the period is 0.40 s:

(i) determine the maximum acceleration; [3]

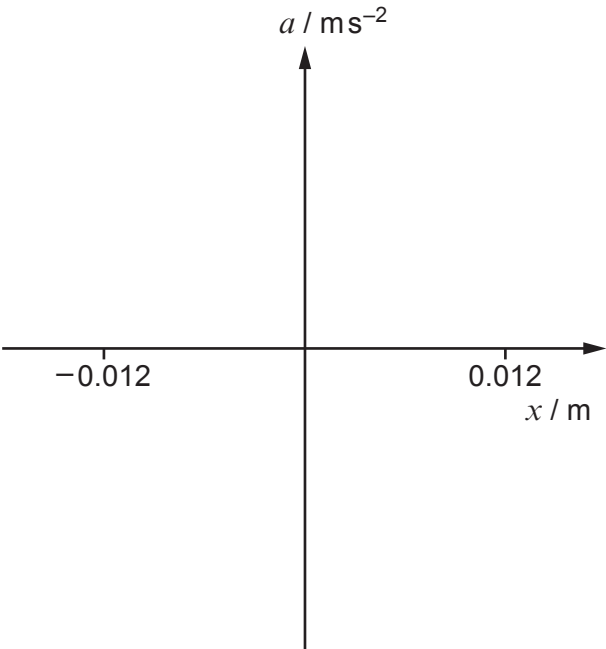
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(ii) sketch a graph showing the variation of a with x . Insert values on the a -axis. Values are given on the x -axis. [3]



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- (c) The oscillating object in part (b) is at its centre position when $t = 0$ and moving in the positive x direction. Insert the missing values in the expression for x in metres. [3]

$$x = \square \cos(\square t + \square)$$

- (d) An oscillating system may be driven by an external force. Describe and explain:

- (i) an application of forced oscillations that is useful; [2]

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- (ii) an example of forced oscillations that should be avoided. [3]

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